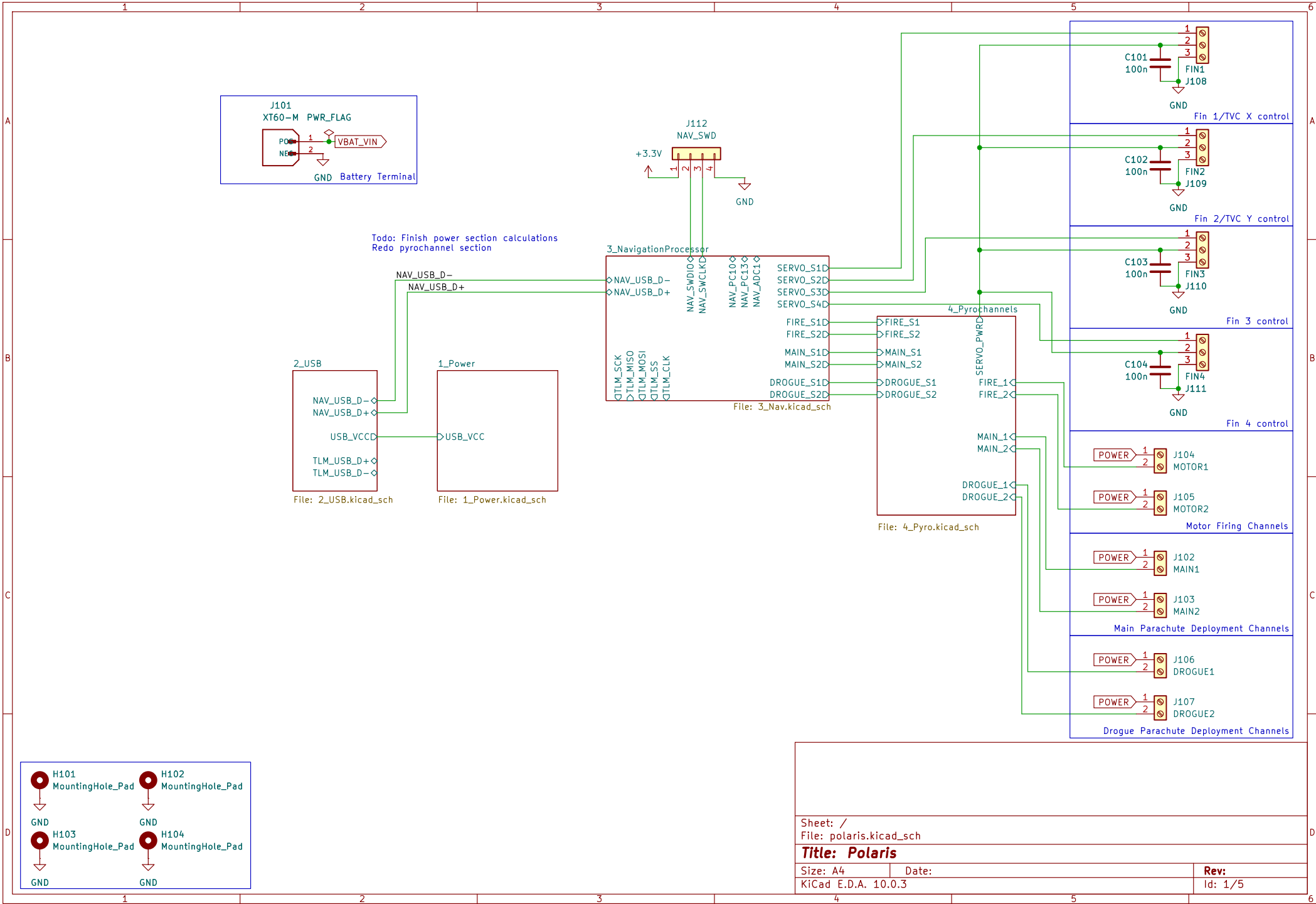
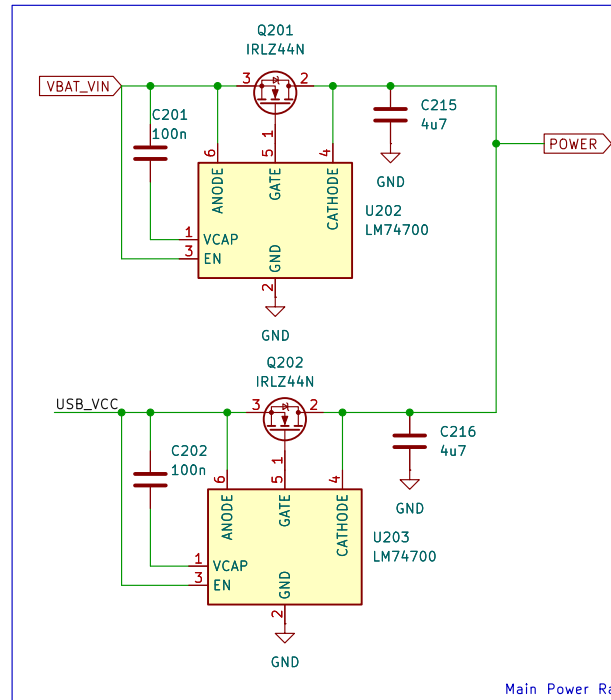
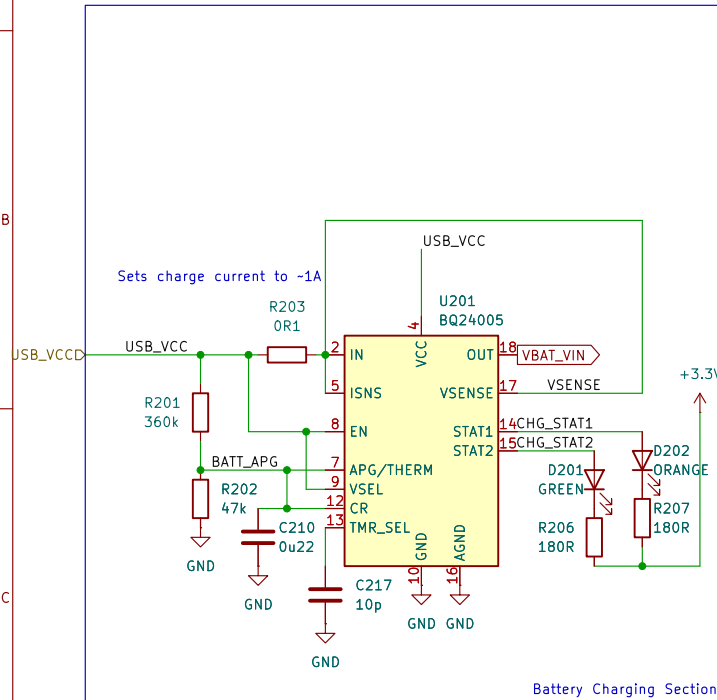




[1] Power



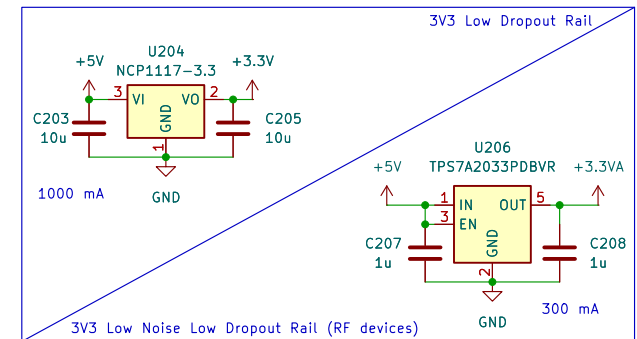
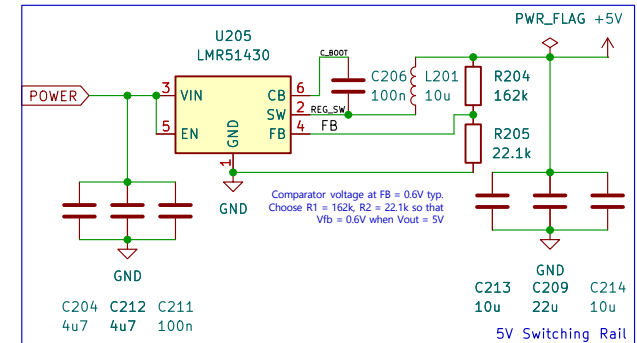
Output inductor calculations:
 $L_{min} = (V_{in,max} - V_{out}) \cdot V_{out} / I_{out} \cdot K \cdot V_{in,max} \cdot f$
 $V_{in,max} = 9V, V_{out} = 5V, I_{out} = 2A, K = 0.3, f = 500kHz$
 $L_{min} = 7.8 \mu H$. Choose nearest available 10 μH

Output capacitor calculations:
 $\Delta V_{out,reg} = K \cdot I_{out} \cdot ESR = 30 mV \text{ max}$
 $= ESR = 0.05 \Omega \text{ max}$

$\Delta V_{out,c} = K \cdot I_{out} / 8 \cdot f \cdot C_{out} = 30 mV \text{ max}$
 $= C_{out} = 5 \mu F$

$C_{out} > 4 \cdot (I_{oh} - I_{ol}) / f \cdot \Delta V_{out,shoot}$
 Using $I_{oh} = 1.5A, I_{ol} = 0.5A, \Delta V_{out,shoot} = 250 mV$
 $C_{out} > 32 \mu F$
 Using $C_{out} > 32 \mu F$ at 50 mΩ ESR
 Choose $C_{out} = 2 \times 10 \mu F + 1 \times 22 \mu F$ at 10V = 42 μF

Please remember to keep the buck converter far from the USB-C receptacle to reduce EMI



Sheet: /1_Power/
 File: 1_Power.kicad_sch

Title: Polaris/Power

Size: A4
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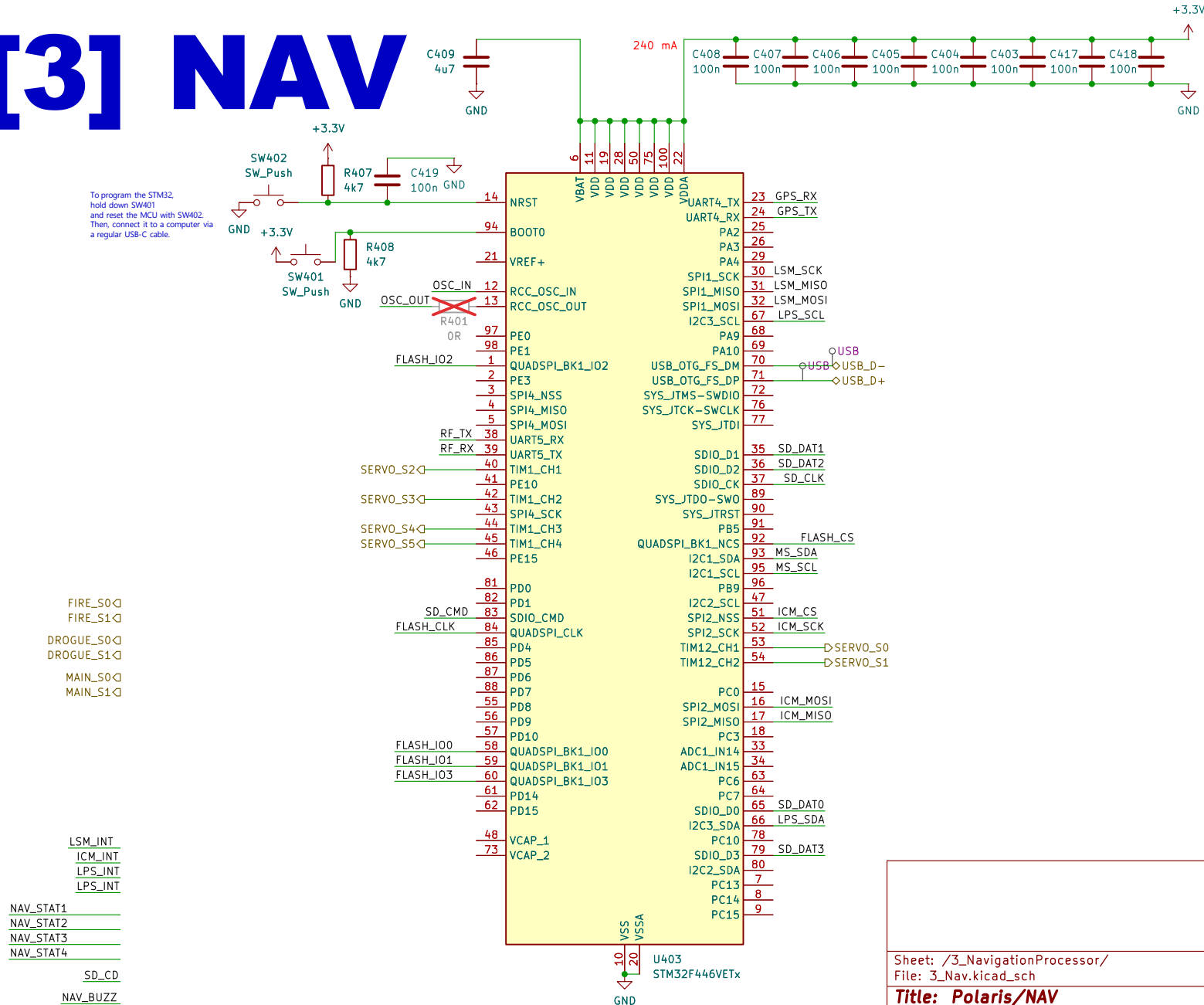


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Id: 3/5

[3] NAV

To program the STM32, hold down SW401 and reset the MCU with SW402. Then, connect it to a computer via a regular USB-C cable.



Sheet: /3_NavigationProcessor/
File: 3_Nav.kicad_sch

Title: Polaris/NAV

Size: A4

Date:

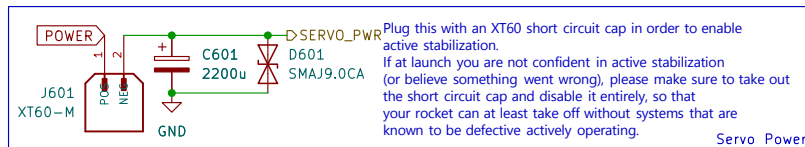
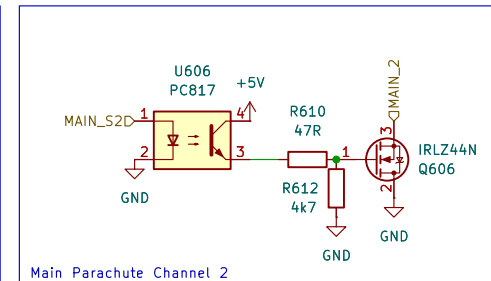
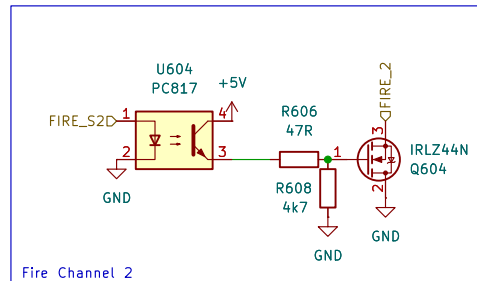
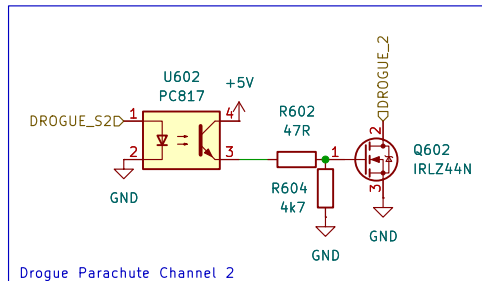
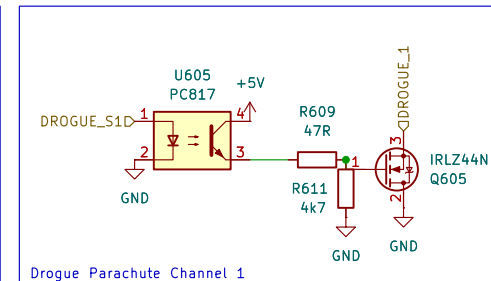
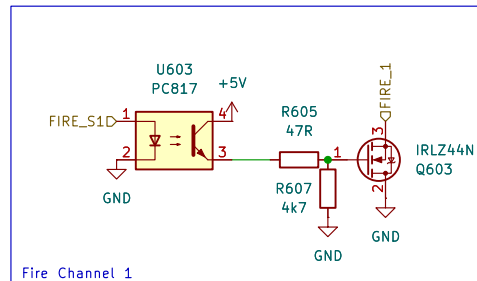
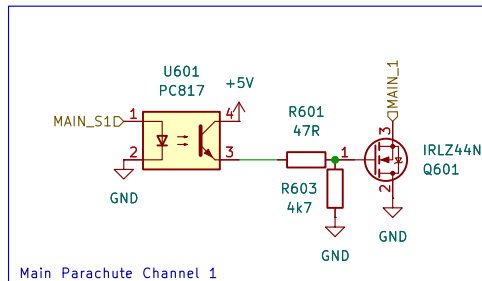
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[5] PYRO

20 mA total



Sheet: /4_Pyrochannels/
File: 4_Pyro.kicad_sch

Title: Polaris/Pyro

Size: A4

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